

DANGQING LIU

☎ (206) 954-1638 📍 Seattle, WA ✉ dangqing@uw.edu

SKILLS & COMPETENCIES

- **Programming Languages:** Python (FastAPI & CV), SystemVerilog, JavaScript, Java, C/C++ (for Arduino)
- **Software & Web Development:** Supabase, UI/UX Design, Git/Linux, MATLAB
- **Embedded & Electronics:** PCB Design (KiCad), FPGA & Modelsim, PID Control, Sensor Integration, Arduino MCU, ESP32-CAM, Fusion 360, Prusa Slicer
- **Languages:** English, Mandarin, German (Basic), Cantonese (Basic)

EDUCATION

University of Washington
Bachelor of Science in Electrical and Computer Engineering

Sep. 2024 - Jun. 2027 (Expected)
Seattle, WA

- **Honors:** ECE Department Honors Candidate, Dean's List in two quarters
- **Relevant Coursework:** Signal Processing, Digital Circuits and Systems, Computer Hardware Skills, Basic Scientific Computing, Advanced Technical Writing.

PUBLICATIONS

- Dangqing Liu, et al. "Personal 3D Printing Visual Dependency in the Generative AI Era: Opportunities and Challenges." *ACM SIGCHI Conference on Designing Interactive Systems (Accepted, In Press)*, 2026.
- Dangqing Liu. "The Current Study and Potential Treatment of Drug Abuse." *Theoretical and Natural Science* 4(1): 316-320, 2023. DOI: 10.54254/2753-8818/4/20220578.

WORK & TEACHING EXPERIENCE

OnlyWorks Inc.
Software Engineering (SWE) Intern

Apr. 2026 – Present
Seattle, WA

- Addressing UI/UX design challenges and developing frontend features using JavaScript to improve user interaction flow and frontend performance.
- Optimizing backend database structures and integrating Supabase to ensure efficient data retrieval and system scalability.

University of Washington (EE 393)
Teaching Assistant

Apr. 2026 – Present
Seattle, WA

- Assisting Prof. Alex Mamishev in EE 393 course operations, grading students' works, guiding students through technical writing and communication concepts.

RESEARCH EXPERIENCE

Sensors, Energy, and Automation Laboratory (SEAL) UW
Undergraduate Research Assistant

Jan. 2026 – Present
Seattle, WA

- Supporting multiple research groups, including a smart sensor integration project, by conducting comprehensive literature reviews and summarizing key findings to accelerate project ideation.
- Assisting researchers in drafting, formatting, and proofreading academic papers to ensure compliance with top-tier conference and journal submission standards.

CRAFT²
Undergraduate Research Assistant

Oct. 2024 – Present
Seattle, WA

- Developed a wearable navigation aid for visually impaired users by integrating haptic feedback motors and force-based directional cues.
- Designed and prototyped the **motor actuation subsystem** using **Arduino Nano** with **motor encoder and DRV8833 motor driver**; implemented real-world haptic response calibration.
- Collaborated on integrating **Google Maps API** and WatchOS radar inputs for real-time obstacle detection and path guidance.
- Iterated hardware design based on user testing to improve device ergonomics and system reliability.

ENGINEERING PROJECTS

HMD Interactive Museum Experience

Mar. 2026 – Present

Algorithm Developer

Seattle, WA

- Developing Python-based computer vision (CV) algorithms to render museum paintings interactive and clickable for Head-Mounted Display (HMD) users.
- Processing dynamic visual inputs to precisely align virtual interactions with physical environments.

Arcade 3-Lane Dodger

Mar. 2026

FPGA Developer

Seattle, WA

- Designed and implemented a retro arcade game on a DE1-SoC FPGA board using SystemVerilog.
- Programmed custom digital logic for real-time collision detection, user input processing, and LED display rendering.

Accessibility-Focused 3D Printing System

Nov. 2025 – Present

Project Lead / Developer

Seattle, WA

- Developing a **Python-based backend** system attached with **OpenAI API** to facilitate remote monitoring and management of 3D printing tasks.
- Implementing **ESP32-CAM** monitoring via motor encoder feedback to optimize the automated printing workflow.
- Building a responsive interface to enhance accessibility and real-time print status visualization.

Line-Tracking Robot

Jun. 2025

Team Lead

Seattle, WA

- Built a fully autonomous robot using **Arduino Mega** and photoresistors, implementing custom **PID-based control logic**.
- Programmed and tuned the PID algorithm for stable dynamic tracking of paths under varying lighting conditions.
- Designed and **3D-modeled** the robot chassis using **Fusion 360** and the Prusa workflow.
- Designed and soldered **custom PCB modules** for sensor integration and power distribution.

EXTRACURRICULAR ACTIVITIES

- **UW SOLAR:** Supported early-stage planning for solar canopy infrastructure at UW parking lots, organizing technical documentation and feasibility evaluations.
- **Personal Interests:** Piano Performance, Composition.

Personal Website: *dangqingliu.github.io*